

Doctorant	Nom prénom / Full name	Quentin NIVON
PhD student	Ecole Doctorale / Doctoral School	MSTII - Mathématiques, Sciences et technologies de l'information, Informatique Spécialité : Informatique
Titre thèse / PhD Title	Analysis, optimisation, and debugging of BPMN processes	
Rapporteur	Nom prénom / Full name	Massimo Mecella
Reviewer	Etablissement / Institution	Sapienza Università di Roma
	Statut, fonction / Status, position	Full professor

Qualité du mémoire, rédaction & illustrations / Thesis quality, style & illustrations (A titre indicatif/For information : Exceptionnel = top 5%, Très bon/very good = top 25 %)

⇒ Satisfaisant / Satisfactory [] Bon / Good [] Très bon / Very good [] Exceptionnel [X]

⇒ Le manuscrit nécessite d'importantes révisions/The manuscript requires major revisions NO

Commentaires/comments :

The thesis addresses a very interesting topic, i.e., how to derive BPMN models from textual specifications, and how to refactor such models in order to consider constraints on resources. These topics are addressed in the research area of Information Systems Engineering (at large), and in conferences like BPM (ranked A in iCORE), ICSSOC (ranked A), EDOC (ranked B), FSE (ranked A*), SEFM (ranked B), etc. Notably, the candidate has already published part of his work in such conferences, including an A* and an A one, and this is particularly appreciable.

Overall the thesis is convincing, well written, easy to follow, with concrete implementations of the tools realizing the techniques, and a robust validation. An interesting aspect is how the thesis, even if adopting Large Language Models (LLMs), which are the killer technology nowadays, does not adopt them in a trivial way but proposes a formal approach to be built on top of them, in order to empower correctness and verifiability of the outcomes. This would make the thesis also ascribable to the novel approach of neurosymbolic AI, or at least this can be presented as an example of what it should mean (personal view of the reviewer). Overall the judgement is therefore excellent.]

Contexte, état de l'art, collaborations / Background, state of the art, collaborations :

Commentaires/comments :

The thesis is rigorously based on the current state-of-the-art in the field, which is very well described and compared. References, to the best of my knowledge, are appropriate and pertinent and up-to-date. The thesis does not demonstrate deep collaborations with other relevant groups (cf. the authors of the main papers, they are mostly the candidate and his supervisor), nevertheless the candidate appears as knowing very well relevant research and notable results in the field, and has attended and published relevant conferences.

I suggest candidate to consider moving chapter 6 after chapter 2, or to incorporate in it.]

Qualité scientifique, méthodologie, expérimentations, validation**Scientific quality, methodology, experiments, validation**

⇒ Satisfaisant / Satisfactory ☐ Bon / Good ☐ Très bon / Very good ☒ Exceptionnel ☐

⇒ *Beaucoup de points majeurs doivent être corrigés / Many major issues need to be corrected* NO

Commentaires/comments :

The scientific quality of the thesis is extremely high, based on an interesting interplay between AI/LLMs, formal methods and model checking, and optimization techniques. The way the candidate merges the concepts in a novel way is particularly appreciable.

An implementation is provided for all proposed techniques, into tools which are available online for large use. A robust validation is performed, presented, and discussed.

It would be interesting if the tools, for repeatability purposes, would be made available in the source code and in some deployment format (a virtual machine, a docker image, etc.) together with instructions for easy replaying. And analogously, it would be interesting to have all the datasets and results available in open repositories, as Zenodo, Kaggle, etc. This is becoming standard in many research communities, and also helps increasing the citations of own works.]

Apports personnels, originalité, valorisation, perspectives**Personal contributions, originality, exploitation and application of results, prospects.***Commentaires/comments :*

The personal contribution of the candidate appears quite high, most of the papers out of the thesis are only with his advisor. The contribution is original, as discussed beforehand, and the application of the results is promising, considering also the implementation level of the proposed techniques.]

Conclusions du rapporteur / Reviewer's conclusions*Commentaires/comments :*

The thesis is excellent, it was a great pleasure to be reviewer of it and to be part of the awarding committee

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Avis du rapporteur / Reviewer's opinion :

Défavorable à la soutenance / Unfavorable to the defence ☐

Favorable ☒

Date 19 November 2025

Signature

Massimo Meccello

Rapport détaillé, commentaires libres, questionnements, corrections demandées

Detailed report, free comments, questions, requested corrections

As the thesis is excellent in my view, I don't have too comments.

Just a few suggestions, as anticipated before:

- Move Chapter 6 after Chapter 2, or incorporate it into Chapter 2
- Make all tools and datasets repeatable]